Products of topological Ramsey spaces

Timothy Trujillo
Texas State University

BLAST 2015
NMSU

1/6/15

timothy.trujillo@txstate.edu

Notation

Suppose that n is a natural number and Y is a set.

Notation

Suppose that n is a natural number and Y is a set.

- $[Y]^n :=$ n-element subsets of Y

Notation

Suppose that n is a natural number and Y is a set.

- $[Y]^n :=$ n-element subsets of Y
- $\omega :=$ The set of natural numbers

Notation

Suppose that n is a natural number and Y is a set.

- $[Y]^n :=$ n-element subsets of Y
- $\omega :=$ The set of natural numbers
- $[\omega]^n :=$ n-element subsets of the natural numbers

Notation

Suppose that n is a natural number and Y is a set.

- $[Y]^n :=$ n-element subsets of Y
- $\omega :=$ The set of natural numbers
- $[\omega]^n :=$ n-element subsets of the natural numbers
- $[\omega]^\omega :=$ infinite subsets of the natural numbers

Notation

Suppose that n is a natural number and Y is a set.

- $[Y]^n :=$ n-element subsets of Y
- $\omega :=$ The set of natural numbers
- $[\omega]^n :=$ n-element subsets of the natural numbers
- $[\omega]^\omega :=$ infinite subsets of the natural numbers
- $[\omega]^{<\omega} :=$ finite subsets of the natural numbers

Notation

Suppose that n is a natural number and Y is a set.

- $[Y]^n :=$ n-element subsets of Y
- $\omega :=$ The set of natural numbers
- $[\omega]^n :=$ n-element subsets of the natural numbers
- $[\omega]^\omega :=$ infinite subsets of the natural numbers
- $[\omega]^{<\omega} :=$ finite subsets of the natural numbers
- $[Y]^\omega :=$ infinite subsets of Y

Notation

Suppose that n is a natural number and Y is a set.

- $[Y]^n :=$ n-element subsets of Y
- $\omega :=$ The set of natural numbers
- $[\omega]^n :=$ n-element subsets of the natural numbers
- $[\omega]^\omega :=$ infinite subsets of the natural numbers
- $[\omega]^{<\omega} :=$ finite subsets of the natural numbers
- $[Y]^\omega :=$ infinite subsets of Y
- $[Y]^{<\omega} :=$ finite subsets of Y

The Ramsey Theorem

Definition

A set $\mathcal{X} \subseteq [\omega]^n$ is **Ramsey** if for each infinite subset X of the natural numbers there exists an infinite set $Y \subseteq X$ such that either $[Y]^n \cap \mathcal{X} = \emptyset$ or $[Y]^n \subseteq \mathcal{X}$.

The Ramsey Theorem

Definition

A set $\mathcal{X} \subseteq [\omega]^n$ is **Ramsey** if for each infinite subset X of the natural numbers there exists an infinite set $Y \subseteq X$ such that either $[Y]^n \cap \mathcal{X} = \emptyset$ or $[Y]^n \subseteq \mathcal{X}$.

Theorem (The Ramsey Theorem)

Suppose n is a natural number. Every subset of $[\omega]^n$ is Ramsey.

The Ramsey Theorem

Definition

A set $\mathcal{X} \subseteq [\omega]^n$ is **Ramsey** if for each infinite subset X of the natural numbers there exists an infinite set $Y \subseteq X$ such that either $[Y]^n \cap \mathcal{X} = \emptyset$ or $[Y]^n \subseteq \mathcal{X}$.

Theorem (The Ramsey Theorem)

Suppose n is a natural number. Every subset of $[\omega]^n$ is Ramsey.

Question

Is every subset of $[\omega]^\omega$ Ramsey?

The Ramsey Theorem

Definition

A set $\mathcal{X} \subseteq [\omega]^n$ is **Ramsey** if for each infinite subset X of the natural numbers there exists an infinite set $Y \subseteq X$ such that either $[Y]^n \cap \mathcal{X} = \emptyset$ or $[Y]^n \subseteq \mathcal{X}$.

Theorem (The Ramsey Theorem)

Suppose n is a natural number. Every subset of $[\omega]^n$ is Ramsey.

Question

Is every subset of $[\omega]^\omega$ Ramsey? No

The Ramsey Property

Definition (The nth approximation map)

Suppose that n is a natural number and $\{x_0, x_1, x_2, \dots\}$ is an infinite subset of natural numbers enumerated in increasing order.

$$r_n(\{x_0, x_1, x_2 \dots\}) = \{x_0, x_1, \dots x_{n-1}\}$$

The Ramsey Property

Definition (The nth approximation map)

Suppose that n is a natural number and $\{x_0, x_1, x_2, \dots\}$ is an infinite subset of natural numbers enumerated in increasing order.

$$r_n(\{x_0, x_1, x_2 \dots\}) = \{x_0, x_1, \dots x_{n-1}\}$$

Definition

For each $X \in [\omega]^\omega$ and each $s \in [\omega]^{<\omega}$, let

$$[s, X] = \{Y \in [\omega]^\omega : \exists n(s = r_n(Y)) \ \& \ Y \subseteq X\}$$

A set $\mathcal{X} \subseteq [\omega]^\omega$ is **Ramsey** if for each $[s, X] \neq \emptyset$ there exists $Y \in [s, X]$ such that either $[s, Y] \subseteq \mathcal{X}$ or $[s, X] \cap \mathcal{X} = \emptyset$.

The Ramsey Property

Definition (The nth approximation map)

Suppose that n is a natural number and $\{x_0, x_1, x_2, \dots\}$ is an infinite subset of natural numbers enumerated in increasing order.

$$r_n(\{x_0, x_1, x_2 \dots\}) = \{x_0, x_1, \dots x_{n-1}\}$$

Definition

For each $X \in [\omega]^\omega$ and each $s \in [\omega]^{<\omega}$, let

$$[s, X] = \{Y \in [\omega]^\omega : \exists n(s = r_n(Y)) \ \& \ Y \subseteq X\}$$

A set $\mathcal{X} \subseteq [\omega]^\omega$ is **Ramsey** if for each $[s, X] \neq \emptyset$ there exists $Y \in [s, X]$ such that either $[s, Y] \subseteq \mathcal{X}$ or $[s, X] \cap \mathcal{X} = \emptyset$.

Theorem (Galvin-Prikry)

The set of Ramsey subsets of $[\omega]^\omega$ forms a σ-algebra of sets.

The Ellentuck Space and the Baire Property

Definition

The **Ellentuck topology** on $[\omega]^\omega$ is the topology generated by the sets $[s, X]$ where $s \in [\omega]^{<\omega}$ and $X \in [\omega]^\omega$.

The Ellentuck Space and the Baire Property

Definition

The **Ellentuck topology** on $[\omega]^\omega$ is the topology generated by the sets $[s, X]$ where $s \in [\omega]^{<\omega}$ and $X \in [\omega]^\omega$.

Definition

A set $\mathcal{X} \subseteq [\omega]^\omega$ has the **Baire property with respect to the Ellentuck topology** if there exists an open set $\mathcal{O}$ with respect to the Ellentuck topology and a meager set $\mathcal{M}$ with respect the Ellentuck topology such that

$$\mathcal{X} = \mathcal{O} \Delta \mathcal{M}.$$

The Ellentuck Space and the Baire Property

Definition

The **Ellentuck topology** on $[\omega]^\omega$ is the topology generated by the sets $[s, X]$ where $s \in [\omega]^{<\omega}$ and $X \in [\omega]^\omega$.

Definition

A set $\mathcal{X} \subseteq [\omega]^\omega$ has the **Baire property with respect to the Ellentuck topology** if there exists an open set $\mathcal{O}$ with respect to the Ellentuck topology and a meager set $\mathcal{M}$ with respect the Ellentuck topology such that

$$\mathcal{X} = \mathcal{O} \Delta \mathcal{M}.$$

Theorem

The collection of subsets of $[\omega]^\omega$ with the Baire property forms a σ-algebra of sets.

The Ellentuck Theorem

Theorem (The Ellentuck Theorem)

A subset of $[\omega]^\omega$ is Ramsey if and only if it has the Baire property with respect to the Ellentuck topology.

The Ellentuck Theorem

Theorem (The Ellentuck Theorem)

A subset of $[\omega]^\omega$ is Ramsey if and only if it has the Baire property with respect to the Ellentuck topology.

Notation

Suppose $n \in \omega$ and $[s, X] \neq \emptyset$.

$$r_n[s, X] = \{r_n(Y) : Y \in [s, X]\}$$

The Ellentuck Theorem

Theorem (The Ellentuck Theorem)

A subset of $[\omega]^\omega$ is Ramsey if and only if it has the Baire property with respect to the Ellentuck topology.

Notation

Suppose $n \in \omega$ and $[s, X] \neq \emptyset$.

$$r_n[s, X] = \{r_n(Y) : Y \in [s, X]\}$$

Corollary

For each $[\emptyset, X] \neq \emptyset$ and each function $f : r_1[\emptyset, X] \to \omega$, there exists $Y \in [\emptyset, X]$ such that either $f \restriction r_1[\emptyset, Y]$ is constant or $f \restriction r_1[\emptyset, Y]$ is strictly increasing.

The Depth Map

Definition

Suppose that $s \in [\omega]^{<\omega}$ and $X \in [\omega]^{\omega}$.

$$\mathrm{depth}_X(s) = \begin{cases} \min\{n : s \subseteq r_n(X)\} & \text{if } s \subseteq X \\ \infty & \text{otherwise} \end{cases}$$

The Depth Map

Definition

Suppose that $s \in [\omega]^{<\omega}$ and $X \in [\omega]^{\omega}$.

$$\mathrm{depth}_X(s) = \begin{cases} \min\{n : s \subseteq r_n(X)\} & \text{if } s \subseteq X \\ \infty & \text{otherwise} \end{cases}$$

Example

Suppose that $E = \{0, 2, 4, 6, 8, \dots\}$, $O = \{1, 3, 5, 7, \dots\}$ and $s \in [\omega]^{<\omega}$.

The Depth Map

Definition

Suppose that $s \in [\omega]^{<\omega}$ and $X \in [\omega]^{\omega}$.

$$\mathrm{depth}_X(s) = \begin{cases} \min\{n : s \subseteq r_n(X)\} & \text{if } s \subseteq X \\ \infty & \text{otherwise} \end{cases}$$

Example

Suppose that $E = \{0, 2, 4, 6, 8, \dots\}$, $O = \{1, 3, 5, 7, \dots\}$ and $s \in [\omega]^{<\omega}$.

- $\mathrm{depth}_\omega(s) = \max(s)$

The Depth Map

Definition

Suppose that $s \in [\omega]^{<\omega}$ and $X \in [\omega]^{\omega}$.

$$\mathrm{depth}_X(s) = \begin{cases} \min\{n : s \subseteq r_n(X)\} & \text{if } s \subseteq X \\ \infty & \text{otherwise} \end{cases}$$

Example

Suppose that $E = \{0, 2, 4, 6, 8, \dots\}$, $O = \{1, 3, 5, 7, \dots\}$ and $s \in [\omega]^{<\omega}$.

- $\mathrm{depth}_\omega(s) = \max(s)$
- $\mathrm{depth}_E\{2, 4, 6\} = 4$

The Depth Map

Definition

Suppose that $s \in [\omega]^{<\omega}$ and $X \in [\omega]^{\omega}$.

$$\mathrm{depth}_X(s) = \begin{cases} \min\{n : s \subseteq r_n(X)\} & \text{if } s \subseteq X \\ \infty & \text{otherwise} \end{cases}$$

Example

Suppose that $E = \{0, 2, 4, 6, 8, \dots\}$, $O = \{1, 3, 5, 7, \dots\}$ and $s \in [\omega]^{<\omega}$.

- $\mathrm{depth}_\omega(s) = \max(s)$
- $\mathrm{depth}_E\{2, 4, 6\} = 4$
- $\mathrm{depth}_O\{2, 4, 6\} = \infty$

The Depth Map

Definition

Suppose that $s \in [\omega]^{<\omega}$ and $X \in [\omega]^{\omega}$.

$$\mathrm{depth}_X(s) = \begin{cases} \min\{n : s \subseteq r_n(X)\} & \text{if } s \subseteq X \\ \infty & \text{otherwise} \end{cases}$$

Example

Suppose that $E = \{0, 2, 4, 6, 8, \dots\}$, $O = \{1, 3, 5, 7, \dots\}$ and $s \in [\omega]^{<\omega}$.

- $\mathrm{depth}_\omega(s) = \max(s)$
- $\mathrm{depth}_E\{2, 4, 6\} = 4$
- $\mathrm{depth}_O\{2, 4, 6\} = \infty$
- $\mathrm{depth}_O\{5\} = 3$

The Ellentuck Topology on $[\omega]^\omega \times [\omega]^\omega$

Definition

Suppose that n is a natural number and $(X, Y) \in [\omega]^\omega \times [\omega]^\omega$.

$$r_n(X, Y) = (r_n(X), r_n(Y))$$

The Ellentuck Topology on $[\omega]^\omega \times [\omega]^\omega$

Definition

Suppose that n is a natural number and $(X, Y) \in [\omega]^\omega \times [\omega]^\omega$.

$$r_n(X, Y) = (r_n(X), r_n(Y))$$

Definition

Suppose $(X, Y), (S, T) \in [\omega]^\omega \times [\omega]^\omega$. $(X, Y) \leq (S, T)$ means that

- $X \subseteq S$, $Y \subseteq T$

The Ellentuck Topology on $[\omega]^\omega \times [\omega]^\omega$

Definition

Suppose that n is a natural number and $(X, Y) \in [\omega]^\omega \times [\omega]^\omega$.

$$r_n(X, Y) = (r_n(X), r_n(Y))$$

Definition

Suppose $(X, Y), (S, T) \in [\omega]^\omega \times [\omega]^\omega$. $(X, Y) \leq (S, T)$ means that

- $X \subseteq S$, $Y \subseteq T$ and
- $\forall n \in \omega, \mathrm{depth}_S r_n(X) \leq \mathrm{depth}_T r_n(Y)$.

The Ellentuck Topology on $[\omega]^\omega \times [\omega]^\omega$

Definition

Suppose that n is a natural number and $(X, Y) \in [\omega]^\omega \times [\omega]^\omega$.

$$r_n(X, Y) = (r_n(X), r_n(Y))$$

Definition

Suppose $(X, Y), (S, T) \in [\omega]^\omega \times [\omega]^\omega$. $(X, Y) \leq (S, T)$ means that

- $X \subseteq S$, $Y \subseteq T$ and
- $\forall n \in \omega, \mathrm{depth}_S r_n(X) \leq \mathrm{depth}_T r_n(Y)$.

Example

Suppose $E = \{0, 2, 4, 6, 8, \dots\}$ and $O = \{1, 3, 5, 7, \dots\}$.

The Ellentuck Topology on $[\omega]^\omega \times [\omega]^\omega$

Definition

Suppose that n is a natural number and $(X, Y) \in [\omega]^\omega \times [\omega]^\omega$.

$$r_n(X, Y) = (r_n(X), r_n(Y))$$

Definition

Suppose $(X, Y), (S, T) \in [\omega]^\omega \times [\omega]^\omega$. $(X, Y) \leq (S, T)$ means that

- $X \subseteq S$, $Y \subseteq T$ and
- $\forall n \in \omega, \mathrm{depth}_S r_n(X) \leq \mathrm{depth}_T r_n(Y)$.

Example

Suppose $E = \{0, 2, 4, 6, 8, \dots\}$ and $O = \{1, 3, 5, 7, \dots\}$.

$$(E, O) \leq (\omega, \omega) \ \ \& \ \ (O, E) \not\leq (\omega, \omega)$$

The Ellentuck Topology on $[\omega]^\omega \times [\omega]^\omega$

Definition (Basic Open Sets)

Let $(X, Y) \in [\omega]^\omega \times [\omega]^\omega$ and $(s, t) \in \bigcup_{n\in\omega}[\omega]^n \times [\omega]^n$.

$$[(s, t), (X, Y)] = \{(S, T) : \exists n((s, t) = r_n(S, T)) \,\&\, (S, T) \leq (X, Y)\}$$

The Ellentuck Topology on $[\omega]^\omega \times [\omega]^\omega$

Definition (Basic Open Sets)

Let $(X, Y) \in [\omega]^\omega \times [\omega]^\omega$ and $(s, t) \in \bigcup_{n\in\omega}[\omega]^n \times [\omega]^n$.

$$[(s, t), (X, Y)] = \{(S, T) : \exists n((s, t) = r_n(S, T)) \,\&\, (S, T) \leq (X, Y)\}$$

Definition (The Ramsey Property)

A set $\mathcal{X} \subseteq [\omega]^\omega \times [\omega]^\omega$ is **Ramsey** if for each $[(s, t), (X, Y)] \neq \emptyset$ there exists $(S, T) \in [(s, t), (X, Y)]$ such that either $[(s, t), (S, T)] \subseteq \mathcal{X}$ or $[(s, t), (S, T)] \cap \mathcal{X} = \emptyset$.

The Ellentuck Topology on $[\omega]^\omega \times [\omega]^\omega$

Definition (Basic Open Sets)

Let $(X, Y) \in [\omega]^\omega \times [\omega]^\omega$ and $(s, t) \in \bigcup_{n\in\omega}[\omega]^n \times [\omega]^n$.

$$[(s, t), (X, Y)] = \{(S, T) : \exists n((s, t) = r_n(S, T)) \,\&\, (S, T) \leq (X, Y)\}$$

Definition (The Ramsey Property)

A set $\mathcal{X} \subseteq [\omega]^\omega \times [\omega]^\omega$ is **Ramsey** if for each $[(s, t), (X, Y)] \neq \emptyset$ there exists $(S, T) \in [(s, t), (X, Y)]$ such that either $[(s, t), (S, T)] \subseteq \mathcal{X}$ or $[(s, t), (S, T)] \cap \mathcal{X} = \emptyset$.

Definition

The **Ellentuck topology** on $[\omega]^\omega \times [\omega]^\omega$ is the topology generated by the sets $[(s, t), (X, Y)]$ where $(s, t) \in \bigcup_{n\in\omega}[\omega]^n \times [\omega]^n$ and $(X, Y) \in [\omega]^\omega \times [\omega]^\omega$.

The Ellentuck Theorem for $[\omega]^\omega \times [\omega]^\omega$

Theorem (T.)

A subset of $[\omega]^\omega \times [\omega]^\omega$ is Ramsey if and only if it has the Baire property with respect to the Ellentuck topology.

The Ellentuck Theorem for $[\omega]^\omega \times [\omega]^\omega$

Theorem (T.)

A subset of $[\omega]^\omega \times [\omega]^\omega$ is Ramsey if and only if it has the Baire property with respect to the Ellentuck topology.

Remark

Let $\pi_0 : r_1[(\emptyset, \emptyset), (\omega, \omega)] \to \omega$ be the map such that $\pi_0(x, y) = x$.

The Ellentuck Theorem for $[\omega]^\omega \times [\omega]^\omega$

Theorem (T.)

A subset of $[\omega]^\omega \times [\omega]^\omega$ is Ramsey if and only if it has the Baire property with respect to the Ellentuck topology.

Remark

Let $\pi_0 : r_1[(\emptyset, \emptyset), (\omega, \omega)] \to \omega$ be the map such that $\pi_0(x, y) = x$. For each $[(\emptyset, \emptyset), (X, Y)] \neq \emptyset$, $\pi_0 \restriction r_1[(\emptyset, \emptyset), (X, Y)]$ is neither strictly increasing nor constant.

The Ellentuck Topology on $\Pi_{i<m}[\omega]^\omega$

Definition

Suppose that $m \in \omega$ and $(X_i)_{i<m}, (Y_i)_{i<m} \in \Pi_{i<m}[\omega]^\omega$.

$$r_n((X_i)_{i<m}) = (r_n(X_i))_{i<m}$$

$(X_i)_{i<m} \leq (Y_i)_{i<m}$ if and only if $\forall i < m$, $X_i \subseteq Y_i$ and

$$\forall i < m, \forall n \in \omega, \mathrm{depth}_{Y_i} r_n(X_i) \leq \mathrm{depth}_{Y_{i+1}} r_n(X_{i+1})$$

The Ellentuck Topology on $\Pi_{i<m}[\omega]^\omega$

Definition

Suppose that $m \in \omega$ and $(X_i)_{i<m}, (Y_i)_{i<m} \in \Pi_{i<m}[\omega]^\omega$.

$$r_n((X_i)_{i<m}) = (r_n(X_i))_{i<m}$$

$(X_i)_{i<m} \leq (Y_i)_{i<m}$ if and only if $\forall i < m$, $X_i \subseteq Y_i$ and

$$\forall i < m, \forall n \in \omega, \mathrm{depth}_{Y_i} r_n(X_i) \leq \mathrm{depth}_{Y_{i+1}} r_n(X_{i+1})$$

Definition (Basic Open Sets)

Let $(X_i)_{i<m} \in \Pi_{i<m}[\omega]^\omega$ and $(s_i)_{i<m} \in \bigcup_{n\in\omega} \Pi_{i<m}[\omega]^n$. Let $[(s_i)_{i<m}, (X_i)_{i<m}]$ denote the set

$$\{(Y_i)_{i<m} : \exists n((s_i)_{i<m} = r_n((Y_i)_{i<m})) \ \& \ (Y_i)_{i<m} \leq (X_i)_{i<m}\}$$

The Ellentuck Theorem on $\Pi_{i<m}[\omega]^\omega$

Definition (The Ramsey Property)

A set $\mathcal{X} \subseteq \Pi_{i<m}[\omega]^\omega$ is **Ramsey** if for each $[(s_i)_{i<m}, (X_i)_{i<m}] \neq \emptyset$ there exists $(Y_i)_{i<m} \in [(s_i)_{i<m}, (X_i)_{i<m}]$ such that either $[(s_i)_{i<m}, (Y_i)_{i<m}] \subseteq \mathcal{X}$ or $[(s_i)_{i<m}, (Y_i)_{i<m}] \cap \mathcal{X} = \emptyset$.

The Ellentuck Theorem on $\Pi_{i<m}[\omega]^\omega$

Definition (The Ramsey Property)

A set $\mathcal{X} \subseteq \Pi_{i<m}[\omega]^\omega$ is **Ramsey** if for each $[(s_i)_{i<m}, (X_i)_{i<m}] \neq \emptyset$ there exists $(Y_i)_{i<m} \in [(s_i)_{i<m}, (X_i)_{i<m}]$ such that either $[(s_i)_{i<m}, (Y_i)_{i<m}] \subseteq \mathcal{X}$ or $[(s_i)_{i<m}, (Y_i)_{i<m}] \cap \mathcal{X} = \emptyset$.

Theorem (T.)

A subset of $\Pi_{i<m}[\omega]^\omega$ is Ramsey if and only if it has the Baire property with respect to the Ellentuck topology.

The Ellentuck Topology on $\Pi_{i\in\omega}[\omega]^\omega$

Definition

Suppose that $(X_i)_{i\in\omega}, (Y_i)_{i\in\omega} \in \Pi_{i\in\omega}[\omega]^\omega$.

$$r_n((X_i)_{i\in\omega}) = (r_n(X_i))_{i<n}$$

$(X_i)_{i\in\omega} \leq (Y_i)_{i\in\omega}$ if and only if $\forall i \in \omega$, $X_i \subseteq Y_i$ and

$$\forall i \in \omega, \forall n \in \omega, \mathrm{depth}_{Y_i} r_n(X_i) \leq \mathrm{depth}_{Y_{i+1}} r_n(X_{i+1})$$

The Ellentuck Topology on $\Pi_{i\in\omega}[\omega]^\omega$

Definition

Suppose that $(X_i)_{i\in\omega}, (Y_i)_{i\in\omega} \in \Pi_{i\in\omega}[\omega]^\omega$.

$$r_n((X_i)_{i\in\omega}) = (r_n(X_i))_{i<n}$$

$(X_i)_{i\in\omega} \leq (Y_i)_{i\in\omega}$ if and only if $\forall i \in \omega$, $X_i \subseteq Y_i$ and

$$\forall i \in \omega, \forall n \in \omega, \mathrm{depth}_{Y_i} r_n(X_i) \leq \mathrm{depth}_{Y_{i+1}} r_n(X_{i+1})$$

Definition (Basic Open Sets)

Let $(X_i)_{i\in\omega} \in \Pi_{i<\omega}[\omega]^\omega$ and $(s_i)_{i<n} \in \bigcup_{m\in\omega} \Pi_{i<m}[\omega]^m$. Let $[(s_i)_{i<n}, (X_i)_{i\in\omega}]$ denote the set

$$\{(Y_i)_{i\in\omega} : \exists n((s_i)_{i<n} = r_n((Y_i)_{i\in\omega})) \ \& \ (Y_i)_{i\in\omega} \leq (X_i)_{i\in\omega}\}$$

The Ellentuck Theorem on $\Pi_{i\in\omega}[\omega]^\omega$

Definition (The Ramsey Property)

A set $\mathcal{X} \subseteq \Pi_{i\in\omega}[\omega]^\omega$ is **Ramsey** if for each $[(s_i)_{i<n}, (X_i)_{i\in\omega}] \neq \emptyset$ there exists $(Y_i)_{i\in\omega} \in [(s_i)_{i<n}, (X_i)_{i\in\omega}]$ such that either $[(s_i)_{i<n}, (Y_i)_{i\in\omega}] \subseteq \mathcal{X}$ or $[(s_i)_{i<n}, (Y_i)_{i\in\omega}] \cap \mathcal{X} = \emptyset$.

The Ellentuck Theorem on $\Pi_{i\in\omega}[\omega]^\omega$

Definition (The Ramsey Property)

A set $\mathcal{X} \subseteq \Pi_{i\in\omega}[\omega]^\omega$ is **Ramsey** if for each $[(s_i)_{i<n}, (X_i)_{i\in\omega}] \neq \emptyset$ there exists $(Y_i)_{i\in\omega} \in [(s_i)_{i<n}, (X_i)_{i\in\omega}]$ such that either $[(s_i)_{i<n}, (Y_i)_{i\in\omega}] \subseteq \mathcal{X}$ or $[(s_i)_{i<n}, (Y_i)_{i\in\omega}] \cap \mathcal{X} = \emptyset$.

Theorem (T.)

A subset of $\Pi_{i\in\omega}[\omega]^\omega$ is Ramsey if and only if it has the Baire property with respect to the Ellentuck topology.

Topological Ramsey Spaces

Definition ([1])

Let $(\mathcal{R}, \leq, r)$ be a triple such that

Definition ([1])

Let $(\mathcal{R}, \leq, r)$ be a triple such that $\mathcal{R}$ is a nonempty set,

Topological Ramsey Spaces

Definition ([1])

Let $(\mathcal{R}, \leq, r)$ be a triple such that $\mathcal{R}$ is a nonempty set, $\leq$ is a quasi-ordering on $\mathcal{R}$ and $r : \mathcal{R} \times \omega \to \mathcal{AR}$.

Topological Ramsey Spaces

Definition ([1])

Let $(\mathcal{R}, \leq, r)$ be a triple such that $\mathcal{R}$ is a nonempty set, $\leq$ is a quasi-ordering on $\mathcal{R}$ and $r : \mathcal{R} \times \omega \to \mathcal{AR}$. For each $a \in \mathcal{AR}$ and each $B \in \mathcal{R}$, let

$$[a, B] = \{A \in \mathcal{R} : A \leq B \ \& \ (\exists n) r_n(A) = a\}.$$

Topological Ramsey Spaces

Definition ([1])

Let $(\mathcal{R}, \leq, r)$ be a triple such that $\mathcal{R}$ is a nonempty set, $\leq$ is a quasi-ordering on $\mathcal{R}$ and $r : \mathcal{R} \times \omega \to \mathcal{AR}$. For each $a \in \mathcal{AR}$ and each $B \in \mathcal{R}$, let

$$[a, B] = \{A \in \mathcal{R} : A \leq B \ \& \ (\exists n) r_n(A) = a\}. \tag{1}$$

The **Ellentuck topology** on $\mathcal{R}$ is the topology generated by the sets $[a, B]$ where $a \in \mathcal{AR}$ and $B \in \mathcal{R}$.

Topological Ramsey Spaces

Definition ([1])

Let $(\mathcal{R}, \leq, r)$ be a triple such that $\mathcal{R}$ is a nonempty set, $\leq$ is a quasi-ordering on $\mathcal{R}$ and $r : \mathcal{R} \times \omega \to \mathcal{AR}$. For each $a \in \mathcal{AR}$ and each $B \in \mathcal{R}$, let

$$[a, B] = \{A \in \mathcal{R} : A \leq B \ \& \ (\exists n) r_n(A) = a\}. \tag{1}$$

The **Ellentuck topology** on $\mathcal{R}$ is the topology generated by the sets $[a, B]$ where $a \in \mathcal{AR}$ and $B \in \mathcal{R}$.

Definition

A subset $\mathcal{X}$ of $\mathcal{R}$ is **Ramsey** if for every $\emptyset \neq [a, A]$, there is a $B \in [a, A]$ such that $[a, B] \subseteq \mathcal{X}$ or $[a, B] \cap \mathcal{X} = \emptyset$.

Topological Ramsey Spaces

Definition ([1])

Let $(\mathcal{R}, \leq, r)$ be a triple such that $\mathcal{R}$ is a nonempty set, $\leq$ is a quasi-ordering on $\mathcal{R}$ and $r : \mathcal{R} \times \omega \to \mathcal{AR}$. For each $a \in \mathcal{AR}$ and each $B \in \mathcal{R}$, let

$$[a, B] = \{A \in \mathcal{R} : A \leq B \ \& \ (\exists n) r_n(A) = a\}. \tag{1}$$

The **Ellentuck topology** on $\mathcal{R}$ is the topology generated by the sets $[a, B]$ where $a \in \mathcal{AR}$ and $B \in \mathcal{R}$.

Definition

A subset $\mathcal{X}$ of $\mathcal{R}$ is **Ramsey** if for every $\emptyset \neq [a, A]$, there is a $B \in [a, A]$ such that $[a, B] \subseteq \mathcal{X}$ or $[a, B] \cap \mathcal{X} = \emptyset$.
A subset $\mathcal{X}$ of $\mathcal{R}$ is **Ramsey null** if for every $\emptyset \neq [a, A]$, there is a $B \in [a, A]$ such that $[a, B] \cap \mathcal{X} = \emptyset$.

Topological Ramsey Spaces

Definition

A triple $(\mathcal{R}, \leq, r)$ with its Ellentuck topology is a **topological Ramsey space** if every subset of $\mathcal{R}$ with the Baire property is Ramsey and if every meager subset of $\mathcal{R}$ is Ramsey null.

Topological Ramsey Spaces

Definition

A triple $(\mathcal{R}, \leq, r)$ with its Ellentuck topology is a **topological Ramsey space** if every subset of $\mathcal{R}$ with the Baire property is Ramsey and if every meager subset of $\mathcal{R}$ is Ramsey null.

Axiom (A.1)

For each $A, B \in \mathcal{R}$,

(a) $r_0(A) = \emptyset$.

(b) $A \neq B$ *implies* $r_i(A) \neq r_i(B)$ *for some* i.

(c) $r_i(A) = r_j(B)$ *implies* $i = j$ *and* $r_k(A) = r_k(B)$ *for all* $k < i$.

Topological Ramsey Spaces

Definition

A triple $(\mathcal{R}, \leq, r)$ with its Ellentuck topology is a **topological Ramsey space** if every subset of $\mathcal{R}$ with the Baire property is Ramsey and if every meager subset of $\mathcal{R}$ is Ramsey null.

Axiom (A.1)

For each $A, B \in \mathcal{R}$,

(a) $r_0(A) = \emptyset$.

(b) $A \neq B$ *implies* $r_i(A) \neq r_i(B)$ *for some* i.

(c) $r_i(A) = r_j(B)$ *implies* $i = j$ *and* $r_k(A) = r_k(B)$ *for all* $k < i$.

Remark

On the basis of this axiom, $\mathcal{R}$ can be identified with a subset of $\mathcal{AR}^\omega$ by associating $A \in \mathcal{R}$ with the sequence $(r_i(A))_{i \in \omega}$.

Axiom (A.2)

There is a quasi-ordering $\leq_{\mathrm{fin}}$ *on* $\mathcal{AR}$ *such that*

(a) $\{a \in \mathcal{AR} : a \leq_{\mathrm{fin}} b\}$ *is finite for all* $b \in \mathcal{AR}$,

(b) $A \leq B$ *iff* $(\forall i)(\exists j)\ r_i(A) \leq_{\mathrm{fin}} r_j(B)$,

(c) $\forall a, b, c \in \mathcal{AR}[a \sqsubseteq b \wedge b \leq_{\mathrm{fin}} c \rightarrow \exists d \sqsubseteq c\ a \leq_{\mathrm{fin}} d]$.

Axiom (A.2)

There is a quasi-ordering $\leq_{\mathrm{fin}}$ on $\mathcal{AR}$ such that

(a) $\{a \in \mathcal{AR} : a \leq_{\mathrm{fin}} b\}$ *is finite for all* $b \in \mathcal{AR}$,

(b) $A \leq B$ *iff* $(\forall i)(\exists j)\ r_i(A) \leq_{\mathrm{fin}} r_j(B)$,

(c) $\forall a, b, c \in \mathcal{AR}[a \sqsubseteq b \wedge b \leq_{\mathrm{fin}} c \rightarrow \exists d \sqsubseteq c\ a \leq_{\mathrm{fin}} d]$.

Notation

For $a \in \mathcal{AR}$ and $B \in \mathcal{R}$, $\mathrm{depth}_B(a)$ is the least i, if it exists, such that $a \leq_{\mathrm{fin}} r_i(B)$. Otherwise we write $\mathrm{depth}_B(a) = \infty$.

Axiom (A.2)

There is a quasi-ordering $\leq_{\mathrm{fin}}$ on $\mathcal{AR}$ such that

(a) $\{a \in \mathcal{AR} : a \leq_{\mathrm{fin}} b\}$ *is finite for all* $b \in \mathcal{AR}$,

(b) $A \leq B$ *iff* $(\forall i)(\exists j)\ r_i(A) \leq_{\mathrm{fin}} r_j(B)$,

(c) $\forall a, b, c \in \mathcal{AR}[a \sqsubseteq b \wedge b \leq_{\mathrm{fin}} c \rightarrow \exists d \sqsubseteq c\ a \leq_{\mathrm{fin}} d]$.

Notation

For $a \in \mathcal{AR}$ and $B \in \mathcal{R}$, $\mathrm{depth}_B(a)$ is the least i, if it exists, such that $a \leq_{\mathrm{fin}} r_i(B)$. Otherwise we write $\mathrm{depth}_B(a) = \infty$.

Axiom (A.3)

For each $A, B \in \mathcal{R}$ *and each* $a \in \mathcal{AR}$,

(a) *If* $\mathrm{depth}_B(a) < \infty$ *then* $[a, A] \neq \emptyset$ *for all* $A \in [\mathrm{depth}_B(a), B]$.

(b) $A \leq B$ *and* $[a, A] \neq \emptyset$ *imply that there is an* $A' \in [\mathrm{depth}_B(a), B]$ *such that* $\emptyset \neq [a, A'] \subseteq [a, A]$.

The Abstract Ellentuck Theorem

Notation

If $n \in \omega$, then $r_n[a, A] = \{r_n(B) : B \in [a, A]\}$

The Abstract Ellentuck Theorem

Notation

If $n \in \omega$, then $r_n[a, A] = \{r_n(B) : B \in [a, A]\}$ and $\mathcal{AR}_n = \{r_n(B) : B \in \mathcal{R}\}$. If $b \in \mathcal{AR}_n$ then we write $|b| = n$.

The Abstract Ellentuck Theorem

Notation

If $n \in \omega$, then $r_n[a, A] = \{r_n(B) : B \in [a, A]\}$ and $\mathcal{AR}_n = \{r_n(B) : B \in \mathcal{R}\}$. If $b \in \mathcal{AR}_n$ then we write $|b| = n$.

Axiom (A.4 - The Pigeonhole Principle for $\mathcal{R}$)

For each $B \in \mathcal{R}$ and each $a \in \mathcal{AR}$, if $\mathrm{depth}_B(a) < \infty$ and $\mathcal{O} \subseteq \mathcal{AR}_{|a|+1}$, then there is $A \in [\mathrm{depth}_B(a), B]$ such that

$$r_{|a|+1}[a, A] \subseteq \mathcal{O} \text{ or } r_{|a|+1}[a, A] \subseteq \mathcal{O}^c. \tag{2}$$

The Abstract Ellentuck Theorem

Notation

If $n \in \omega$, then $r_n[a, A] = \{r_n(B) : B \in [a, A]\}$ and $\mathcal{AR}_n = \{r_n(B) : B \in \mathcal{R}\}$. If $b \in \mathcal{AR}_n$ then we write $|b| = n$.

Axiom (A.4 - The Pigeonhole Principle for $\mathcal{R}$)

For each $B \in \mathcal{R}$ and each $a \in \mathcal{AR}$, if $\mathrm{depth}_B(a) < \infty$ and $\mathcal{O} \subseteq \mathcal{AR}_{|a|+1}$, then there is $A \in [\mathrm{depth}_B(a), B]$ such that

$$r_{|a|+1}[a, A] \subseteq \mathcal{O} \text{ or } r_{|a|+1}[a, A] \subseteq \mathcal{O}^c. \tag{2}$$

Theorem (The Abstract Ellentuck Theorem, [1])

If $(\mathcal{R}, \leq, r)$ is a closed subspace of $\mathcal{AR}^\omega$ and satisfies **A.1, A.2, A.3** *and* **A.4** *then $(\mathcal{R}, \leq, r)$ forms a topological Ramsey space.*

Definition $((\mathcal{R} \times \mathcal{S}, \leq, r))$

Suppose that $(\mathcal{R}, \leq, r)$ and $(\mathcal{S}, \leq, r)$ are topological Ramsey spaces satisfying **A.1**-**A.4**

Definition $((\mathcal{R} \times \mathcal{S}, \leq, r))$

Suppose that $(\mathcal{R}, \leq, r)$ and $(\mathcal{S}, \leq, r)$ are topological Ramsey spaces satisfying **A.1**-**A.4** and $(X, Y), (S, T) \in \mathcal{R} \times \mathcal{S}$.

Definition $((\mathcal{R} \times \mathcal{S}, \leq, r))$

Suppose that $(\mathcal{R}, \leq, r)$ and $(\mathcal{S}, \leq, r)$ are topological Ramsey spaces satisfying **A.1**-**A.4** and $(X, Y), (S, T) \in \mathcal{R} \times \mathcal{S}$.

$$r_n(X, Y) = (r_n(X), r_n(Y))$$

Definition $((\mathcal{R} \times \mathcal{S}, \leq, r))$

Suppose that $(\mathcal{R}, \leq, r)$ and $(\mathcal{S}, \leq, r)$ are topological Ramsey spaces satisfying **A.1-A.4** and $(X, Y), (S, T) \in \mathcal{R} \times \mathcal{S}$.

$$r_n(X, Y) = (r_n(X), r_n(Y))$$

$(X, Y) \leq (S, T)$ means that

- $X \leq S$, $Y \leq T$

Definition $((\mathcal{R} \times \mathcal{S}, \leq, r))$

Suppose that $(\mathcal{R}, \leq, r)$ and $(\mathcal{S}, \leq, r)$ are topological Ramsey spaces satisfying **A.1-A.4** and $(X, Y), (S, T) \in \mathcal{R} \times \mathcal{S}$.

$$r_n(X, Y) = (r_n(X), r_n(Y))$$

$(X, Y) \leq (S, T)$ means that

- $X \leq S$, $Y \leq T$ and
- $\forall n \in \omega, \mathrm{depth}_S r_n(X) \leq \mathrm{depth}_T r_n(Y)$.

Definition $((\mathcal{R} \times \mathcal{S}, \leq, r))$

Suppose that $(\mathcal{R}, \leq, r)$ and $(\mathcal{S}, \leq, r)$ are topological Ramsey spaces satisfying **A.1**-**A.4** and $(X, Y), (S, T) \in \mathcal{R} \times \mathcal{S}$.

$$r_n(X, Y) = (r_n(X), r_n(Y))$$

$(X, Y) \leq (S, T)$ means that

- $X \leq S$, $Y \leq T$ and
- $\forall n \in \omega, \mathrm{depth}_S r_n(X) \leq \mathrm{depth}_T r_n(Y)$.

Lemma (T.)

Suppose that $(\mathcal{R}, \leq, r)$ and $(\mathcal{S}, \leq, r)$ are topological Ramsey spaces satisfying **A.1**-**A.4**. *$(\mathcal{R} \times \mathcal{S}, \leq, r)$ satisfies* **A.1** - **A.3**.

Definition $((\mathcal{R} \times \mathcal{S}, \leq, r))$

Suppose that $(\mathcal{R}, \leq, r)$ and $(\mathcal{S}, \leq, r)$ are topological Ramsey spaces satisfying **A.1**-**A.4** and $(X, Y), (S, T) \in \mathcal{R} \times \mathcal{S}$.

$$r_n(X, Y) = (r_n(X), r_n(Y))$$

$(X, Y) \leq (S, T)$ means that

- $X \leq S$, $Y \leq T$ and
- $\forall n \in \omega, \mathrm{depth}_S r_n(X) \leq \mathrm{depth}_T r_n(Y)$.

Lemma (T.)

Suppose that $(\mathcal{R}, \leq, r)$ and $(\mathcal{S}, \leq, r)$ are topological Ramsey spaces satisfying **A.1**-**A.4**. *$(\mathcal{R} \times \mathcal{S}, \leq, r)$ satisfies* **A.1** - **A.3**.

Question

What additional assumptions, if any, must be made about $(\mathcal{R}, \leq, r)$ and $(\mathcal{S}, \leq, r)$ so that **A.4** holds for $(\mathcal{R} \times \mathcal{S}, \leq, r)$?

Thank You!

[1] Stevo Todorcevic.
Introduction to Ramsey Spaces (AM-174).
Princeton University Press, 2010.